

# ADVANCED QUANTUM MECHANICS

## Exercise 9 – presentation session – 09.12.2025



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### 1. Fermi pressure

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Consider  $N$  identical fermions confined in a one-dimensional infinite potential well of length  $L$ . The single-particle stationary states are

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}, \quad n = 1, 2, \dots$$

- (a) For  $N = 2$ , write the properly antisymmetrized two-particle ground-state wavefunction built from the orbitals  $\phi_1(x)$  and  $\phi_2(x)$ .
- (b) Show that the corresponding ground-state energy is

$$E_{\text{GS}}^{(2)} = E_1 + E_2 = \frac{\hbar^2 \pi^2}{2mL^2} (1^2 + 2^2).$$

Comment briefly why both particles cannot occupy  $\phi_1(x)$ .

By induction, the total energy of the  $N$ -fermion ground state is therefore  $E_{\text{tot}}(L) = \sum_{n=1}^N E_n$

- (i) Compute the pressure in one dimension defined as

$$P(L) = - \frac{\partial E_{\text{tot}}}{\partial L}.$$

- (ii) Find the large  $N$  limit of the pressure.
- (iii) Interpret the result physically by noting that the pressure is strictly positive and persists even at zero temperature. Explain why forcing fermions into a smaller length  $L$  increases the energy via Pauli exclusion, and thus generates a mechanical pressure.

**Comment:** This effect is seen in

- **White dwarfs:** electron Fermi pressure stabilizes the star against gravitational collapse.
- **Neutron stars:** neutron Fermi pressure (plus interaction corrections) provides the dominant support.
- **Cold atom Fermi gases:** the cloud expands faster than a Bose gas because of Fermi pressure.
- **Metals:** conduction electrons possess a large zero-temperature kinetic energy that produces Fermi pressure.

## 2. The two-site spinless Hubbard model

Consider spinless fermions on two sites, labeled  $\ell$  (left) and  $r$  (right). There is at most one fermion per site due to the Pauli principle. We introduce annihilation operators  $\hat{c}_\ell, \hat{c}_r$  and corresponding number operators  $\hat{n}_\ell = \hat{c}_\ell^\dagger \hat{c}_\ell, \hat{n}_r = \hat{c}_r^\dagger \hat{c}_r$ .

A minimal interacting model (often called the spinless Hubbard or extended Hubbard model) is

$$\hat{H} = -t \left( \hat{c}_\ell^\dagger \hat{c}_r + \hat{c}_r^\dagger \hat{c}_\ell \right) + V \hat{n}_\ell \hat{n}_r,$$

where  $t > 0$  is the hopping amplitude between the two sites and  $V$  is a nearest-neighbor interaction energy when both sites are occupied.

We work in the Fock basis  $|n_\ell n_r\rangle$  with  $n_\ell, n_r \in \{0, 1\}$ .

(i) **Basis and particle-number sectors.** List all four basis states explicitly:

$$|00\rangle, |10\rangle, |01\rangle, |11\rangle,$$

and group them according to the total particle number  $N = n_\ell + n_r$ . Identify the sectors  $N = 0, N = 1$ , and  $N = 2$ .

(ii) **One-particle sector: bonding and antibonding states.** Restrict  $\hat{H}$  to the  $N = 1$  sector spanned by  $|10\rangle$  and  $|01\rangle$ .

(a) Act with  $\hat{H}$  on  $|10\rangle$  and  $|01\rangle$  and show that, in the basis  $\{|10\rangle, |01\rangle\}$ , the Hamiltonian has the matrix representation

$$H^{(N=1)} = \begin{pmatrix} 0 & -t \\ -t & 0 \end{pmatrix}.$$

(b) Diagonalize this  $2 \times 2$  matrix and find the eigenstates

$$|+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle),$$

with corresponding eigenenergies. Identify which is the bonding (lower energy) and which is the antibonding (higher energy) combination.

(iii) **Two-particle sector and interaction energy.** Consider now the  $N = 2$  state  $|11\rangle$ .

(a) Show that the hopping term  $-t(\hat{c}_\ell^\dagger \hat{c}_r + \hat{c}_r^\dagger \hat{c}_\ell)$  annihilates  $|11\rangle$  because Pauli exclusion forbids putting two spinless fermions on the same site.

(b) Evaluate  $\hat{n}_\ell \hat{n}_r |11\rangle$  and show that  $|11\rangle$  is an eigenstate of  $\hat{H}$  with energy  $V$ .

(iv) **Summary and interpretation.** Summarize the full spectrum of  $\hat{H}$  in all particle-number sectors and comment briefly on the competition between kinetic energy (hopping) and interaction energy  $V$  when going from  $N = 1$  to  $N = 2$ .

## 3. Commutators leading to the Gross–Pitaevskii equation

Consider a dilute Bose gas in one dimension described by bosonic field operators

$$[\hat{\Psi}(x), \hat{\Psi}^\dagger(x')] = \delta(x - x'), \quad [\hat{\Psi}(x), \hat{\Psi}(x')] = 0.$$

The second-quantized Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}},$$

with single-particle part

$$\hat{H}_0 = \int dx \hat{\Psi}^\dagger(x) \left( -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x),$$

and contact interactions

$$\hat{H}_{\text{int}} = \frac{g_{1D}}{2} \int dx \hat{\Psi}^\dagger(x) \hat{\Psi}^\dagger(x) \hat{\Psi}(x) \hat{\Psi}(x).$$

(a) **Single-particle part.** Show explicitly that

$$[\hat{\Psi}(x), \hat{H}_0] = \left( -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x).$$

*Hint:* Rewrite the kinetic term in symmetric form,

$$\hat{H}_{\text{kin}} = \int dx' \frac{\hbar^2}{2m} \partial_{x'} \hat{\Psi}^\dagger(x') \partial_{x'} \hat{\Psi}(x'),$$

and use

$$[\hat{\Psi}(x), \partial_{x'} \hat{\Psi}^\dagger(x')] = \partial_{x'} \delta(x - x').$$

Evaluate

$$\int dx' \partial_{x'} \delta(x - x') \partial_{x'} \hat{\Psi}(x')$$

by integrating by parts and assuming vanishing boundary terms.

(b) **Interaction part.** Compute the commutator

$$[\hat{\Psi}(x), \hat{H}_{\text{int}}],$$

starting from

$$\hat{H}_{\text{int}} = \frac{g_{1D}}{2} \int dx' \hat{\Psi}^\dagger(x') \hat{\Psi}^\dagger(x') \hat{\Psi}(x') \hat{\Psi}(x').$$

*Hints:*

- Use the product identity

$$[A, BC] = [A, B]C + B[A, C].$$

Show that the operator product produces

$$[\hat{\Psi}(x), \hat{\Psi}^\dagger(x') \hat{\Psi}^\dagger(x') \hat{\Psi}(x') \hat{\Psi}(x')] = 2 \delta(x - x') \hat{\Psi}^\dagger(x') \hat{\Psi}(x') \hat{\Psi}(x').$$

Conclude that

$$[\hat{\Psi}(x), \hat{H}_{\text{int}}] = g_{1D} \hat{\Psi}^\dagger(x) \hat{\Psi}(x) \hat{\Psi}(x).$$

(c) **Final operator equation.** Combine (a) and (b) to obtain the full Heisenberg equation

$$i\hbar \partial_t \hat{\Psi}(x, t) = \left( -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x, t) + g_{1D} \hat{\Psi}^\dagger(x, t) \hat{\Psi}(x, t) \hat{\Psi}(x, t),$$

which forms the basis for deriving the Gross–Pitaevskii equation via the mean-field substitution  $\hat{\Psi}(x, t) \rightarrow \psi(x, t)$ .